UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Graphs and diagrams

Every graph and diagram in scope, redrawn as clean vector figures.

Class 011 · Lagrange's method of undetermined multipliers · compiled 15 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S011·5	Deck 011, slide 5 (rendered .pptx)
T011a 00:01:08	Transcript “Lagrange's method of undetermined coefficient” (theory lecture), segment start
T011b 00:05:03	Transcript “Problems on Lagrange's method of undetermined multipliers”
CW U1 P33	Unit 1 class-work problem 33
Extra Q57	Extra question list “lagrange-ques”, question 57
ESA Jul-23 1c	Past paper: Jul 2023, question 1.c

Contents

1. Figure G1.1 · Why the method works: a level curve touches the constraint
2. Figure G2.1 · The soap bubble: $x^2 - z^2 = 1$ nearest the origin
3. Figure G2.2 · $5x^2 + 6xy + 5y^2 = 8$ between the circles of radius 1 and 2

Colours follow the Okabe–Ito order: blue = the constraint curve, vermilion and green = level curves of the function being optimised, grey = guides. Figure G1.1 is added for clarity (why the method works). Figure G2.1 redraws the deck's “soap bubble” picture of slide 011·7 as a cross-section; Figure G2.2 draws the ESA curve $5x^2 + 6xy + 5y^2 = 8$ with its nearest and farthest points.

1 Lagrange's method: the auxiliary function and the procedure

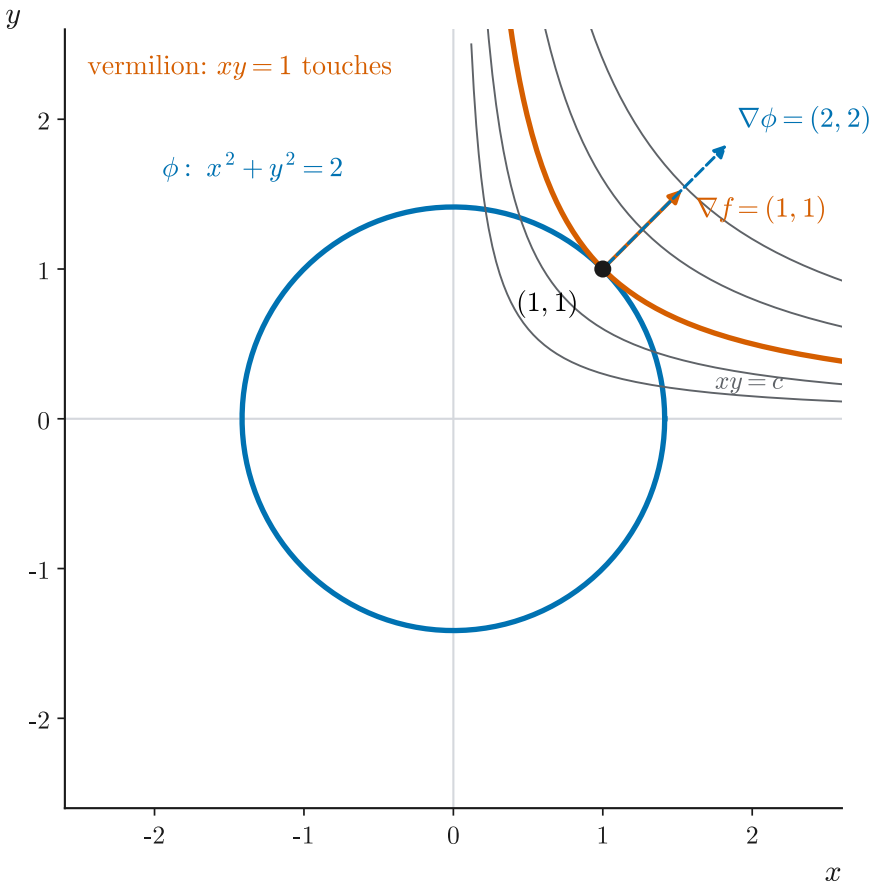


Figure G1.1. Why the method works: a level curve touches the constraint. ESA ★

ADDED FOR CLARITY S011·3 · S011·7

How to read it. Maximise $f = xy$ on the circle $x^2 + y^2 = 2$ (blue). The level curves $xy = c$ (grey hyperbolas) cross the circle until $c = 1$, where $xy = 1$ (vermilion) just touches it at $(1, 1)$. There ∇f and $\nabla\phi$ point the same way (arrows): $\nabla f + \lambda\nabla\phi = 0$ with $\lambda = -\frac{1}{2}$.

2 Geometric problems: inscribed solids, nearest and farthest points

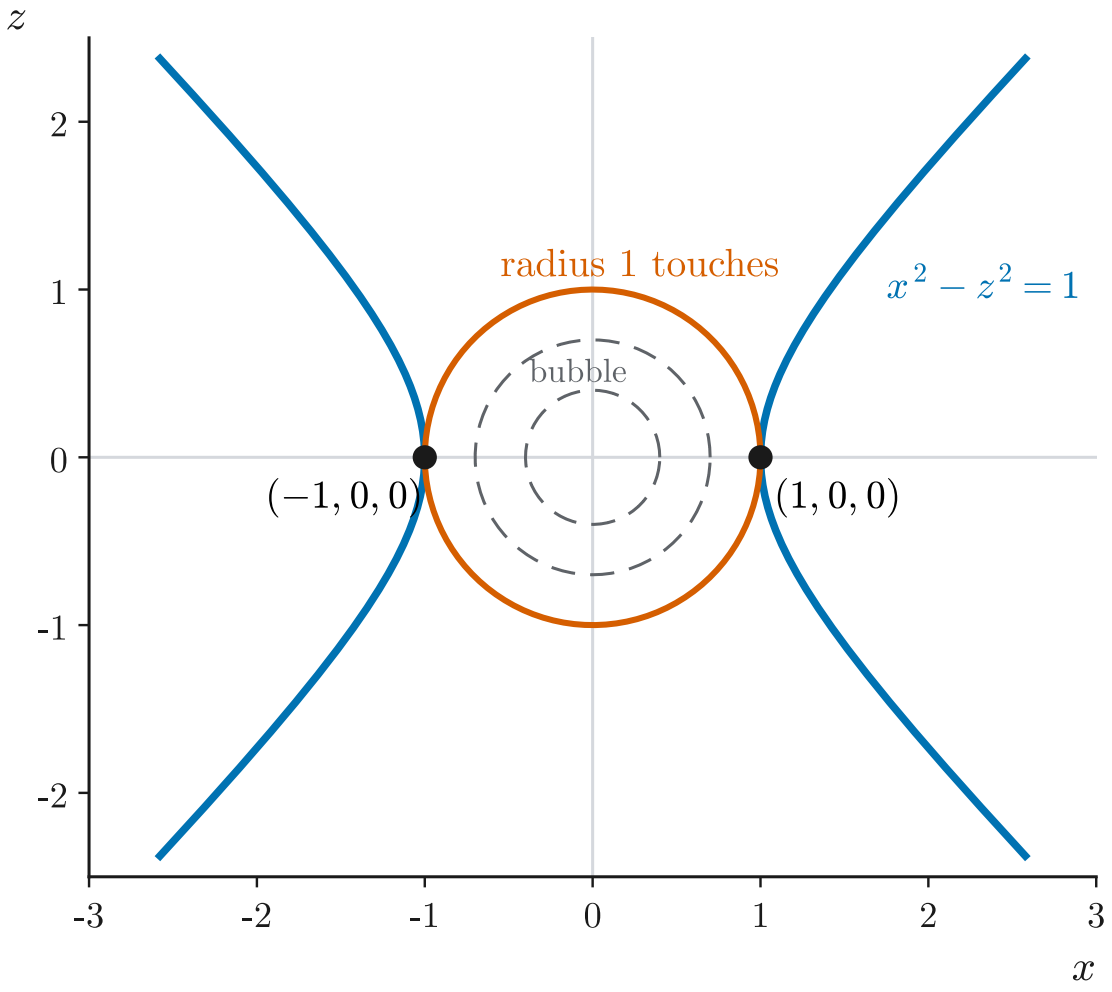


Figure G2.1. The soap bubble: $x^2 - z^2 = 1$ nearest the origin. ESA ★★

S011·7 · S011·9

How to read it. Cross-section $y = 0$. The hyperbolic cylinder appears as the hyperbola $x^2 - z^2 = 1$ (blue). Circles centred at the origin (grey, growing like a soap bubble) first touch it at $(\pm 1, 0)$ with radius 1 (vermilion): the nearest points $(\pm 1, 0, 0)$.

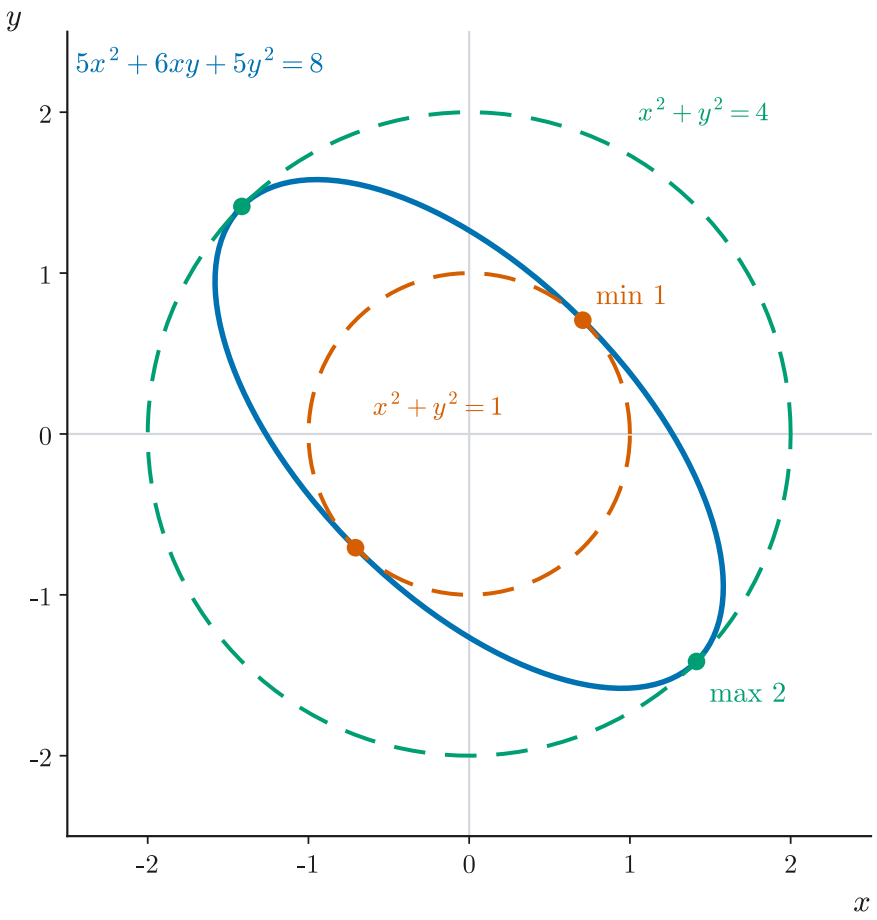


Figure G2.2. $5x^2 + 6xy + 5y^2 = 8$ between the circles of radius 1 and 2. ESA ★★★

ADDED FOR CLARITY ESA Jul-23 1c

How to read it. The curve is an ellipse (blue) tilted at $45°$. The circle $x^2 + y^2 = 1$ (vermilion) touches it from inside at $\pm(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$: minimum distance 1. The circle $x^2 + y^2 = 4$ (green) touches it from outside at $\pm(\sqrt{2}, -\sqrt{2})$: maximum distance 2.